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Thermodynamic modeling of Au–Ce–Sn ternary system

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ABSTRACT

Phase relationships in Au–Ce–Sn ternary system have been thermodynamically modeled by using the CALPHAD technique. Out of the three binary systems, two systems (i.e. Au–Ce, and Sn–Ce) were thermodynamically assessed in this work, and the description of the third one (i.e. Au–Sn) was significantly improved from that of the previous one by adopting the recently published lattice stability of Sn(hcp). *Ab initio* calculations were employed to support the present thermodynamic assessment via calculating the enthalpies of formation of the relevant binary compounds as well as those of the two stoichiometric ternary compounds. All the intermetallic compounds in binary systems and two out of the seven ternary compounds were treated as stoichiometric phases. The other five ternary compounds were modeled with two or three sublattice sites based upon their homogeneity ranges. The solution phases, including liquid, fcc, bcc, hcp and dhcp, were modeled as substitutional solution phases. The Au–Ce–Sn ternary system was assessed by utilizing the thermodynamic descriptions of the three binary systems, and by taking into account the recently reported isothermal section at 750 °C. The set of thermodynamic parameters, obtained in this work, can be used to reproduce the experimentally determined phase equilibria at 750 °C. Besides, the liquidus projection was extrapolated utilizing the present database.

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1. Introduction

Bonding of dissimilar materials is one of the essential steps in the fabrication of modern electronic devices. Prominent examples include integrated circuit (IC) and Micro-Electro-Mechanical Systems (MEMS) packaging technologies. In wafer-level MEMS packaging the bonding systems play a key role in determining the reliability of the device during operation. This is true no matter whether “dielectric bonding” or “metallic bonding” techniques are used. Recently, metallic bonding has been drawing a growing interest in MEMS packaging field owing to its distinguished advantages coming from the use of metal layers. With metallic bonding applying, the bonding layers can simultaneously provide not only the required mechanical supporting but also electrical and thermal connections. Therefore, it is essential to obtain a consistent thermodynamic description of the various metal bonding systems, thus gaining better understanding on the evolution of the microstructure of the interconnections during fabrication, in reliability testing and under operation.

Among those various metal bonding systems, Au–Sn system seems to be receiving most of the attention at the moment owing to its feasible mechanical and physical properties. It can be used as an Au–20 wt%Sn eutectic solder [1], but also as bonding metallization in Transient Liquid Phase (TLP) wafer-level bonding [2]. However, the application of Au–Sn bonding system in wafer-level bonding is limited due to several reasons: (a) the high cost; (b) poor bondability on many surfaces when encountering the wafer-level bonding without interlayers implanted (i.e. surfaces of nitrides, carbides, oxides, silicon and GaAs) [3,4]; and (c) formation of tin oxides during deposition [3,4]. It has been suggested and demonstrated that by alloying the Au–Sn with rare earth (RE) elements these problems could be solved due to the following reasons: (a) the RE alloying elements are safely stored in micron-scale, finely-dispersed intermetallic islands within the solder matrix, and when released, cause the chemical reactions at the interface thus producing strong bonds; and (b) RE metals show a stronger affinity for oxygen than most metals and tend to reduce metal oxides owing to the formation of RE-oxides. Among the 15 lanthanide elements, Ce is one of the most popular candidates for this purpose [5]. Therefore, in the present work, the thermodynamic assessment of the Au–Ce–Sn system was carried out by combining *ab initio* techniques [6,7] with the CALPHAD approach [8]. As a result, a set of self-consistent thermodynamic parameters were obtained.

The paper is organized as follows: a brief review of previous work is presented in Section 2. The applied methods and models that were used will be described in Section 3. The results obtained and the related discussions can be found from Section 4.

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Finally, conclusions drawn from this simulation are summarized in Section 5.

2. A brief review of previous works

2.1. The Au–Ce binary system

The first version of Au–Ce phase diagram was experimentally determined by Vogel et al. [9]. In that work, four compounds were reported, i.e. $AuCe_2$, AuCe, Au_2Ce and Au_3Ce phases, as well as the invariant reactions involved in these phases. However, the Ce, used by Vogel et al. [9], was reported to be 98–99% pure and melted at 777 °C, around 21 °C lower than the present accepted value. Just as clarified in [10], the impure nature in rare earth metals would result in an incorrect diagram; besides, the experimental data published in [9] were just superimposed in the diagram thus leading to a higher risk for bringing inaccuracy data when citing these values. Then, in 1971, Mcmasters et al. [11] investigated the crystal structures of the compounds in Au–Ce system, except for Au_6Ce. They found that the previously accepted Au_3Ce phase [9,12] turned out to be as the $Au_{3.6}Ce$ compound due to the misindexing of their powder patterns [12]. In the present work, the commonly accepted $Au_{51}Ce_{14}$ formula is taken to express this phase, as recommended by Gschneidner et al. [13,14]. Besides, the melting point of βAuCe, and the isotropic transition temperature of βAuCe→αAuCe were also reported by Mcmasters et al. [11]. As for the crystal structure of Au_6Ce, it was reported with the prototype of Au_6Pr in 1974 [15]. Later, the phase diagram information of the Au–Ce binary system was systematically reviewed by Gschneidner et al. in 1986 and 1987. [13,14] The crystal structures of those intermetallic compounds were summarized in the latter

Table 1
Crystal structure of binary compounds in Au–Ce, and Ce–Sn binary systems.

Phase	Pearson symbol	Space group	Prototype	Lattice parameters, nm			Refs.
				a	*b*	*c*	
Au	cF4	Fm3m	Cu	0.40784	…	…	[14]
αCe	cF4	Fm3m	Cu	0.485	…	…	
βCe	hP4	$P6_3/mmc$	αLa	0.36810	…	1.1857	
γCe	cF4	Fm3m	Cu	0.51610	…	…	
δCe	cI2	Im3m	W	0.412	…	…	
$AuCe_2$	oP12	Pnma	Co_2Si	0.7271 ± 0.7	0.5073 ± 0.6	0.932 ± 0.9	
αAuCe	oP8	Pnma	FeB	0.7430	0.4660	0.5930	
βAuCe	oC8	Cmcm	CrB	0.3900	1.114	0.4750	
Au_4Ce_3	hR42	…	Pu_3Pd_4	1.3887	…	0.617	[17]
Au_2Ce	oI12	Imma	$CeCu_2$	0.4528	0.7203	0.8068	[14]
$Au_{51}Ce_{14}$	hP68	P6/m	$Gd_{14}Ag_{51}$	1.281 1.277	… …	0.9327 ± 0.3 0.929	
Au_6Ce	mC28	C2/c	$PrAu_6$	0.78 ± 1	0.778 ± 1	0.908 ± 1	
βSn	tI4	$I4_1/amd$		0.5829		0.318	[22]
αSn	cF8	$Fd\overline{3}m$					
Ce_3Sn	cP4	$Pm\overline{3}m$	$AuCu_3$	0.4929 0.493 0.4927 0.4926	… … … …	… … … …	[23]
βCe_5Sn_3	hP16	$P6_3/mcm$	Mn_5Si_3	0.933	…	0.679	[22]
αCe_5Sn_3	tI32	I4/mcm	W_5Si_3	1.2591 1.26 1.2599 1.258	… … … …	0.6174 0.617 0.6179 0.6153	[23]
Ce_5Sn_4	oP36	Pnma	Sm_5Ge_4	0.8337 0.8348 0.836	1.605 1.603 1.6074	0.848 0.8492 0.8484	
$Ce_{11}Sn_{10}$	tI84	I4/mmm	$Ho_{11}Ge_{10}$	1.22 1.97 1.206–1.212	… … …	1.79 1.782 1.781–1.795	
Ce_3Sn_5	oC32	Cmcm	Pu_3Pd_5	1.0255 1.0243	0.8225 0.8214	1.058 1.0652	
Ce_3Sn_7	oC28	…	Ce_3Sn_7	0.4524	2.5742	0.461	
Ce_2Sn_5	oC28	…	Ce_2Sn_5	0.4559 0.459	3.5014 3.514	0.4619 0.466	
$CeSn_3$	cP4	$Pm\overline{3}m$	$AuCu_3$	0.4721 0.4724	… …	… …	

work. Besides, they also proposed a eutectoid reaction of Au_2Ce at 465 K [14]. A complete phase diagram of the Au–Ce system, from 300 °C to 1500 °C, was proposed in [13,14], excluding two uncertain invariant temperatures involving the Au_6Ce phase. In 1990, Gschneidner et al. [16] briefly introduced the experimentally determined Au–Ce phase diagram again [9,11]. The proposed phase diagram was almost identical to the one published in 1987 [14], with the exception of the composition of most of the eutectic reactions, and the decomposition of Au_2Ce at low temperature [16]. Later, in 2001, a new phase (namely, Au_4Ce_3) was reported by Ferro et al. [17]. However, no evidence of the Au_4Ce_3 was shown in the investigation of an isothermal section of Au–Ce–Sn ternary system at 750 °C (1023 K) by Boulet et al. Moreover, they found that the Au_6Ce phase was unstable above 1023 K [18]. Therefore, in this work, the phase diagram of Au–Ce system is reassessed by taking into account these experimentally determined phase information [9,11,17,18]. For example, in this work, some changes are made, comparing with the work of [13,14,16]: (1) the Au_4Ce_3 phase, formed lower than 1023 K, is included; [17,18] and (2) the formation temperature of Au_6Ce is treated as lower than 1023 K [18]; and accordingly, a lower temperature was proposed for the reaction temperature of the eutectic reaction involving Au_6Ce by comparing with those shown in [13,14,16]. There are seven compounds and eleven invariant reactions in this system, as listed in Tables 1 and 2, respectively.

Regarding the thermochemical information of the Au–Ce system, only a limited amount of data can be found in literature [17]. Fitzner et al. investigated the enthalpy of mixing at 1473 ± 2 K, and the enthalpies of formation of Au_2Ce, AuCe and $Au_{51}Ce_{14}$ at 298 K, by using direct synthesis calorimetry [19]. Later, the enthalpy of

Table 2
Invariant reactions in Au–Ce binary system, comparing with experimental data [9,11,18] and assessed values [13,14,16].

Reaction	Composition (at% Au)			T/K	Refs.
$L \rightarrow \gamma Ce + AuCe_2$	~16.0	–	–	793	[9]
	~16.0	–	–	~793	[13]
	16.0	–	–	793	[14]
	~16.0	–	–	~793	[16]
	16.4	–	–	793	This work
$L + \beta AuCe \rightarrow AuCe_2$	~215	–	–	930	[9]
	~23.5	–	–	938	[13]
	25.2	–	–	938	[14]
	~24	–	–	938	[16]
	27.7	–	–	940	This work
$L \rightarrow \beta AuCe$	50.0	–	–	1623	[9]
	50.0	–	–	1645 ± 10	[11]
	50.0	–	–	1645 ± 10	[13]
	50.0	–	–	1645	[14]
	50.0	–	–	1645 ± 10	[16]
	50.0	–	–	1645	This work
$\beta AuCe \rightarrow \alpha AuCe$	–	–	–	753 ± 5	[11]
	–	–	–	753 ± 10	[13]
	–	–	–	753 ± 10	[16]
	–	–	–	753	This work
$L \rightarrow Au_2Ce + \beta AuCe$	~62.0	–	–	1268	[9]
	≈61.3	–	–	1268	[13]
	62.8	–	–	1353	[14]
	61.3	–	–	1268	[16]
	61.5	–	–	1337	This work
$L \rightarrow Au_2Ce$	66.7	–	–	1403	[9]
	66.7	–		1403	[13]
	66.7	–	–	1403	[14]
	66.7			1403	[16]
	66.7	–	–	1403	This work
$L \rightarrow Au_2Ce + Au_{51}Ce_{14}$	~69.0	–	–	1298	[9]
	~69.0	–	–	1298	[13]
	72.0	–	–	1298	[14]
	~69	–	–	1298	[16]
	72.4	–	–	1311	This work
$L \rightarrow Au_{51}Ce_{14}$	75.0	–	–	~1423	[9]
	79.7	–	–	1423	[13]
	79.7	–	–	1423	[14]
	79.7	–	–	1423	[16]
	79.7	–	–	1423	This work
$L + Au_{51}Ce_{14} \rightarrow Au_6Ce$	~89	–	–	1143[a]	[13]
	87.4	–	–	1143	[14]
	~89	–	–	?	[16]
	–	–	–	< 1023	[18]
	89.2	–	–	1021	This work
$L \rightarrow Au_6Ce + Au(fcc)$	~90.2	–	–	1081	[9]
	~90.5	–	–	1081[a]	[13]
	91	–	–	1081	[14]
	~90.5	–	–	~1081	[16]
	–	–	–	< 1023	[18]
	91.7	–	–	941	This work
$\beta AuCe + Au_2Ce \rightarrow Au_4Ce_3$	57.1	–	–	< 1023	[18]
	57.1	–	–	998	This work
$Au_2Ce \rightarrow Au_4Ce_3 + Au_{51}Ce_{14}$	66.7	–	–	465	[14]
	–	–	–	–	This work

[a] Uncertain temperature.

Table 3
Invariant reactions in Ce–Sn binary system.

Reaction	Composition (at% Au)			T/K	Refs.
$L \rightarrow \delta Ce + Ce_3Sn$	~4.0	~1	–	998	[21]
	…[a]	…	–	1002	[23]
	4.2	1.1	–	999	This work
$\delta Ce \rightarrow \gamma Ce + Ce_3Sn$	~2	–	–	963	[21]
	…	–	–	975	[23]
	0.9	–	–	973	This work
$L + \beta Ce_5Sn_3 \rightarrow Ce_3Sn$	…	–	–	1216	[21]
	~13	–	–	1214	This work
$L + Ce_5Sn_4 \rightarrow \beta Ce_5Sn_3$	…	–	–	1778	[21]
$L \rightarrow Ce_5Sn_4 + \beta Ce_5Sn_3$*	40	–	–	1770	This work
$L \rightarrow Ce_5Sn_3$*	37.5	–	–	1781	This work
$\beta Ce_5Sn_3 \rightarrow \alpha Ce_5Sn_3$	–	–	–	…	[21]
	–	–	–	> 1023	[18]
	–	–	–	…	This work
$L \rightarrow Ce_5Sn_4$	44.4	–	–	1783	[21]
	44.4	–	–	1788	[22]
	44.4	–	–	1788	This work
$L + Ce_5Sn_4 \rightarrow Ce_{11}Sn_{10}$	…	–	–	~1673	[21]
	…	–	–	~1648[b]	[22]
	55.3	–	–	1635	This work
$L \rightarrow Ce_{11}Sn_{10} + Ce_3Sn_5$	58.2	–	–	1428 ± 10	[22]
	61.3	–	–	1451	This work
$L \rightarrow Ce_3Sn_5$	62.5	–	–	1453 ± 10	[22]
	62.5	–	–	1453	This work
$L \rightarrow Ce_3Sn_7 + Ce_3Sn_5$	66.5	–	–	1403 ± 10	[22]
	68.0	–	–	1405	This work
$L + Ce_2Sn_5 \rightarrow Ce_3Sn_7$	…	–	–	1408 ± 10	[22]
	68.4	–	–	1407	This work
$L + CeSn_3 \rightarrow Ce_2Sn_5$	…	–	–	1418 ± 10	[22]
	71.1	–	–	1420	This work
$L \rightarrow CeSn_3$	75	–	–	1443 ± 10	[22]
	75	–	–	1442	This work
$L \rightarrow CeSn_3 + Sn$	99.5	–	–	498 ± 5	[22]
	99.1	–	–	498	This work

[a] Indicates that no data are available.
[b] Proposed data without determination.

formation of another phase (i.e. $AuCe_2$) at 300 K was measured with the high temperature calorimetric technique [17].

2.2. The Ce–Sn binary system

The first Ce–Sn phase diagram, reported by Vogel [20] in 1911, was obtained by using thermal and metallographic analyses. This diagram is considered as the first while incorrect rare earth binary phase diagram. Later, in 1943, Vogel et al. continued to investigate this system by focusing on the 75–100 at%Sn rich part [9]. In that work, a partial phase diagram (~50–92 at%Sn) was published, with the comparison with the work of [20]. However, as discussed in the last section, the phase diagram information in [9,20] is given smaller weight in the present assessment due to accuracy issues. Then, in 1988 [21], a new partial phase diagram of this system was reported, which covered the part from 0 to 50 at%Sn. It was determined by utilizing differential thermal analysis (DTA). The crystal structures of the intermetallic phases determined by X-ray powder and single-crystal diffraction analyses were also reported in [21]. At the same time the remaining part of the Ce–Sn phase diagram (i.e. from 50 at%Sn to pure Sn) was published [22]. Again the DTA method was adopted for phase boundary detection. However, no information about the isotropic transition regarding the βCe_5Sn_3 (high temperature) and αCe_5Sn_3 (low temperature phase) was included. In 1998, Riani et al. summarized all the available information of the Ce–Sn system including the phase diagram information, crystal structures of the IMCs (as shown in Table 1), and thermochemical data [23]. Several differences were found in [19] in comparison with the results from [21] and [22], for instance, the two invariant temperatures in Ce-rich region (listed in Table 3) and the homogeneity range for the $Ce_{11}Sn_{10}$ phase. However, the homogeneity range of the $Ce_{11}Sn_{10}$ phase was not experimentally detected in the isothermal section of the Au–Ce–Sn system at 1023 K [18]. The αCe_5Sn_3 was observed to be stable at 1023 K, while the transition temperature for the isotropic transition, $\beta Ce_5Sn_3 \rightarrow \alpha Ce_5Sn_3$, remained unclear [18]. Therefore, in the present simulation, the $Ce_{11}Sn_{10}$ was treated as a line compound, and the isotropic transition ($\beta Ce_5Sn_3 \rightarrow \alpha Ce_5Sn_3$) was ignored.

As mentioned above the thermodynamic data for Ce–Sn system are summarized in [23]. Therefore, it will not be repeated here. All the relevant thermochemistry data are listed in Table 4.

2.3. The Au–Sn binary system

The binary Au–Sn system has been intensively investigated [24–27]. Among all of these assessment works, the parameters published in 1988 [24] did not take the SGTE unary compilation; thus that work is eliminated to extrapolate multi-components system. The shortage of the work of Liu et al. [25] was clarified by Grolier et al. [26] and Dong et al. [27]. They both made an improvement from [25]: Grolier et al. [26] reassessed the Au–Sn system by considering more experimental information, and the work of Dong et al. [27] made a modification on the basis of the work of Liu et al. by employing the *ab initio* approach to calculate the enthalpy of formation of these line compounds in Au–Sn system. However, recently the lattice stability data for Sn(hcp) were modified by SGTE (i.e. $3900-7.646T+G_{Sn}^{SER}$ instead of $3900-4.4T+G_{Sn}^{SER}$). Hence, an incorrect binary diagram of Au–Sn system would be obtained, when combining these formerly reported thermodynamic parameters [26,27] (i.e. assessed by using original unary data) with the new revised unary data from SGTE. Besides, in 2010, Dębski et al. measured the enthalpies of formation for several compounds in Sn-rich side [28]. Therefore, the thermodynamic description of this system is required to be updated accordingly.

2.4. The Au–Ce–Sn ternary system

For the Au–Ce–Sn ternary system, there is very little information available either on the phase equilibria, or on the thermodynamics of the system. Only one isothermal section at 750 °C has

Table 4
Enthalpy of formations of compounds in Au–Ce–Sn ternary systems: *ab initio* calculated results at 0 K, experimentally determined data at around 298 K, and assessment outcomes via the present database; Reference states: Au (Fcc), Ce (Fcc), and Sn (Bct); unit: kJ/mol.

Phase	*ab initio* kJ/mol	Fitzner [19], 298 K[a]	Ferro [17], 300 K[a]	Borzone [34], 300 K[a]	Palenzona [35], 298 K [a]	Colinet [37], 300 K[b]	Borsese [36], 300 K [a]	This work 289 K
$AuCe_2$	−50.99	…	−55.0 ± 1	…	…	…	…	−59.39
βAuCe	−74.71	…	…	…	…	…	…	…
αAuCe	−75.36	−62.8 ± 4.2	…	…	…	…	…	−76.02
Au_4Ce_3	−74.32	…	…	…	…	…	…	−73.82
Au_2Ce	−71.46	−57.5 ± 4.7	…	…	…	…	…	−70.00
$Au_{51}Ce_{14}$	…	−49.5 ± 2.2	…	…	…	…	…	−57.71
Au_6Ce	−39.76	…	…	…	…	…	…	−41.59
Ce_3Sn	−43.82	…	…	−49.37 ± 2.1	…	−53	…	−53.72
βCe_5Sn_3	−65.34	…	…	…	…	…	…	−70.03[c]
αCe_5Sn_3	−65.89	…	…	−73.22 ± 4.2	…	−75	−75.3 ± 4.2	…
Ce_5Sn_4	−73.25	…	…	−83.68 ± 4.2	…	−83	…	−73.46
$Ce_{11}Sn_{10}$	−73.85	…	…	…	…	…	…	−72.21
Ce_3Sn_5	−66.98	…	…	−75.31 ± 4.2	…	−80	…	−65.38
Ce_3Sn_7	−60.02	…	…	…	…	…	…	−62.19
Ce_2Sn_5	−58.15	…	…	…	…	…	…	−61.29
$CeSn_3$	−51.64	…	…	−58.58 ± 2.1	−49.1 ± 6%	−58	…	−57.62
τ_1	…	…	…	…	…	…	…	−41.85 to −50.89
τ_3	…	…	…	…	…	…	…	−79.00
τ_2	…	…	…	…	…	…	…	−70.38 to −85.61
τ_4	−80.99	…	…	…	…	…	…	−86.46
τ_5	−71.26	…	…	…	…	…	…	−74.77
τ_6	…	…	…	…	…	…	…	−70.00
τ_7	…	…	…	…	…	…	…	−71.89

[a] Experimentally determined by calorimetry.
[b] Theoretically calculated via Miedema's semiempirical model.
[c] Without considering the isotropic transition in this work.

been experimentally determined by using optical microscopy, electron probe micro-analyzer (EPMA), and X-ray powder analysis [18]. It was found that many of the binary phases showed marked ternary solubilities. In the same investigation, a total of 7 ternary compounds were observed. Their crystal structures have also been reported in [18], as listed in Table 6. In addition, all ternary experimental information is listed in Table 7.

3. Calculation methods

In this work, the thermodynamic description was carried on, with the help of *ab initio* calculations. The *ab initio* approach was employed to calculate the enthalpy of formation of binary compounds in Au–Ce (except that for $Au_{51}Ce_{14}$) and Ce–Sn systems, as well as the enthalpies of formation of two stoichiometric ternary compounds (i.e. τ_4 and τ_5, in Table 4).

Table 5
Invariant reactions in Au–Sn system, with a comparison to [26,27].

Reaction	Composition (at% Sn)			T/K	Ref.
L+Fcc(Au)→$Au_{10}Sn$	20.9	7.0	8.7	805.7	[26]
	20.5	7.2	8.9	805.4	[27]
	20.5	7.3	8.9	805.7	This work
L+$Au_{10}Sn$→Hcp	21.0	9.2	10.9	794.5	[26]
	21.0	9.2	11.1	794.5	[27]
	21.0	9.2	11.2	794.5	This work
L→Hcp+AuSn	31.1	16.0	–	553.3	[26]
	30.2	16.3	–	556.4	[27]
	30.3	16.2	–	555.6	This work
Hcp+AuSn→Au_5Sn	13.6	–	–	463	[26]
	13.7	–	–	452	[27]
	13.7	–	–	451.6	This work
L→AuSn	–			691.7	[26]
	–			691.4	[27]
	–			691.8	This work
L+AuSn→$AuSn_2$	71.0 –		–	582.6	[26]
	73.3	–	–	584.6	[27]
	73.3	50.8	–	584.3	This work
L+$AuSn_2$→$AuSn_4$	86.4	–	–	524.4	[26]
	86.6	–	–	525.3	[27]
	87.1	–	–	525.3	This work
L→$AuSn_4$+Bct(Sn)	94.8	–	–	488.6	[26]
	94.2	–	–	483.1	[27]
	94.6	–	–	484.5	This work
$AuSn_4$→$AuSn_2$+Bct(Sn)	–			322.7	[27]
	–			323.1	This work
$Au_{10}Sn$→Fcc(Au)+Hcp	5.9	1.9	8.0	300.5	[27]
$Au_{10}Sn$→Fcc(Au)+Hcp	5.9	1.6	8.0	301.3	This work
Hcp→Au_5Sn+Fcc(Au)	8.0	–	1.6	274.3	[27]
	8.0	–	1.3	275.6	This work
Au_5Sn→Fcc (Au)+AuSn	–	1.0	–	248.7	[27]
	–	0.9	–	249.1	This work

3.1. Ab initio calculations

This approach, as implemented in the highly-efficient Vienna *ab initio* simulation package (VASP) [6,7], is performed by using the scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) pseudopotential formalism [29,30]. The exchange correlation energy has been computed using the generalized gradient approximation (GGA) [31]. A constant plane-wave energy cutoff (i.e. 500 eV) was employed to study the enthalpies of formation of binary compounds in Au–Ce (except for $Au_{51}Ce_{14}$) and Ce–Sn systems and the two ternary compounds. Brillouin zone integrations are performed by combining the Monkhorst–Pack k-point meshes [32] and the Methfessel–Paxton technique [33] with a smearing parameter of 0.2 eV. The reciprocal space (k-point) meshes are increased to achieve convergence with a precision of 1 meV/atom. The total energy is converged numerically to less than 1 × 10-6 eV/unit with respect to electronic, ionic and unit cell degrees of freedom, and the latter two are relaxed using calculated forces with a preconditioned conjugated gradient algorithm. After structural optimization, calculated forces are converged to less than 0.01 eV/Å. All calculations are performed using the "high" precision setting within the VASP input file to avoid wrap-around errors.

The enthalpies of formation of the IMCs are calculated by the following equation:

$$\Delta H(A_xB_y) = E_{total}(A_xB_y) - xE_{total}(A) - yE_{total}(B) \quad (1)$$

where $E_{total}(A_xB_y)$, $E_{total}(A)$, and $E_{total}(B)$ are the calculated total energies (per atom at T=0 K) of the IMCs as well as those for pure A and B, respectively.

3.2. Thermodynamic model

The lattice stability of the element i (i=Au, Ce and Sn) is referred to as the constant enthalpy value of the standard element reference H_i^{SER} at 298.15 K and 1 bar as recommended by SGTE (Scientific Group Thermodata Europe, http://www.sgte.org/). Different models were employed to describe the solution phases and intermetallic compounds. The details of the relevant models are given below.

3.2.1. Solution phases: liquid, Ce(bcc) and Ce(fcc)

Conventional substitutional solution model was applied to describe the liquid, fcc, bcc, hcp and dhcp solutions. The molar Gibbs energy of a solution phase Φ (Φ=liquid, fcc, bcc, hcp and dhcp) can be represented as a sum of the Gibbs energy for these pure components, the ideal entropy term describing a random mixing of components, and the excess Gibbs energy describing the deviation from ideal

Table 6
Crystal structure information of ternary compounds in Au–Ce–Sn system [18].

Phase	Space group/Pearson symbol/Prototype	Lattice parameters, nm		
		a	b	c
τ_1: Au_5CeSn to $Au_{13}Ce_3Sn_4$	$I23$, cI168-filled $Be_{17}Ru_3$-type	1.5096(2) 1.5127(3)	…	…
τ_2: $Au_{2-x}CeSn_x$	Hexagonal, intergrowth of AlB_2-type	0.8313(1)	…	0.9517(1)
τ_3: Au_2CeSn	Hexagonal, hP16, superstructure of the $ZrPt_2Al$-type	0.9517(1)	…	0.9591(3)
τ_4: AuCeSn	$P6_3$/mcm, hP6, CaIn2-type	0.4731(2)	…	0.7712(3)
τ_5: $AuCe_5Sn_3$	$P6_3$/mcm, hP18, Hf_5CuSn_3-type	0.9645(5)	…	0.6754(4)
τ_6: Au_2CeSn_2	Orthorhombic, superstructure of the $ThCr_2Si_2$-type	0.4575(1)	1.359(1)	1.0176(1)
τ_7: $Au_3Ce_2Sn_4$	Tetragonal, superstructure of the $ThCr_2Si_2$-type	0.9142(2)	…	1.0359(3)

Table 7
Experimental information of compounds in Au–Ce–Sn ternary system at 750 °C, with the comparison to the assessed results.

Phase	Determined solubility or homogeneity ranges [18]	assessed results
AuCe	9 at%Sn	9 at%Sn
Au_2Ce	9 at%Sn	9 at%Sn
$Au_{51}Ce_{14}$	7 at%Sn	7 at%Sn
Ce_3Sn	10 at%Au	10 at%Au
Ce_5Sn_3	4 at%Au	4 at%Au
Ce_5Sn_4	4 at%Au	4 at%Au
Ce_3Sn_5	5 at%Au	5 at%Au
$CeSn_3$	10 at%Au	9.79 at%Au
τ_1	$Au_{0.714}Ce_{0.143}Sn_{0.143}$-$Au_{0.65}Ce_{0.15}Sn_{0.2}$	$Au_{0.714}Ce_{0.143}Sn_{0.143}$–$Au_{0.65}Ce_{0.15}Sn_{0.2}$
τ_2	$Au_{0.5}Ce_{0.333}Sn_{0.167}$-$Au_{0.446}Ce_{0.333}Sn_{0.22}$	$Au_{0.5}Ce_{0.333}Sn_{0.167}$–$Au_{0.446}Ce_{0.333}Sn_{0.22}$
τ_3	$Au_{0.5}Ce_{0.25}Sn_{0.25}$- $Au_{0.525}Ce_{0.25}Sn_{0.225}$	$Au_{0.5}Ce_{0.25}Sn_{0.25}$–$Au_{0.525}Ce_{0.25}Sn_{0.225}$
τ_4	line compound: AuCeSn	AuCeSn
τ_5	line compound: $AuCe_5Sn_3$	$AuCe_5Sn_3$
τ_6	$Au_{0.4}Ce_{0.2}Sn_{0.4}$ towards τ_7	$Au_{0.4}Ce_{0.2}Sn_{0.4}$–$Au_{0.382}Ce_{0.2}Sn_{0.418}$
τ_7	$Au_{0.333}Ce_{0.222}Sn_{0.445}$ towards τ_6	$Au_{0.333}Ce_{0.222}Sn_{0.445}$–$Au_{0.345}Ce_{0.222}Sn_{0.433}$

behavior, i.e.,

$$G^{\phi} = \sum_{i=\mathrm{Au,Ce,Sn}} x_i {}^0G_i^{\phi} + RT \sum_{i=\mathrm{Au,Ce,Sn}} x_i \ln(x_i) + {}^{\mathrm{ex}}G^{\phi} \tag{2}$$

where R is the gas constant, T is the temperature in Kelvin, x_i is the molar fraction of component i(i=Au, Ce and Sn), and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i, with the reference state of ϕ. Here the excess term is formulated as

$$\begin{aligned} {}^{\mathrm{ex}}G^{\phi} = {} & x_{\mathrm{Au}}x_{\mathrm{Ce}} \sum_{j=0,1\ldots}^{N} {}^{(j)}L_{\mathrm{Au,Ce}}^{\phi}(x_{\mathrm{Au}}-x_{\mathrm{Ce}})^j \\ & + x_{\mathrm{Au}}x_{\mathrm{Sn}} \sum_{j=0,1\ldots}^{N} {}^{(j)}L_{\mathrm{Au,Sn}}^{\phi}(x_{\mathrm{Au}}-x_{\mathrm{Sn}})^j \\ & + x_{\mathrm{Ce}}x_{\mathrm{Sn}} \sum_{j=0,1\ldots}^{N} {}^{(j)}L_{\mathrm{Ce,Sn}}^{\phi}(x_{\mathrm{Ce}}-x_{\mathrm{Sn}})^j + x_{\mathrm{Au}}x_{\mathrm{Ce}}x_{\mathrm{Sn}}L_{\mathrm{Au,Ce,Sn}}^{\phi} \end{aligned} \tag{3}$$

where ${}^{(j)}L_{\mathrm{Au,Sn}}^{\phi}$ is cited from our previous work [27] and the interaction parameters ${}^{(j)}L_{\mathrm{Au,Ce}}^{\phi}$ and ${}^{(j)}L_{\mathrm{Ce,Sn}}^{\phi}$ were taken to have the following temperature dependence:

$${}^{(j)}L^{\phi} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} \tag{4}$$

Parameters a and b were optimized in this work. The ternary interaction parameters $L_{\mathrm{Au,Ce,Sn}}^{\phi}$ were not optimized here, and were set to be zero due to the lack of reliable experimental information.

3.2.2. Binary compounds

Significant solid-solution range for the binary IMCs has not been reported in binary phase diagram. The only exception is $Ce_{11}Sn_{10}$. However, the binary solubility range of $Ce_{11}Sn_{10}$ phase was not confirmed in the experimental investigations of the Au–Ce–Sn ternary system [18]. Thus, all the binary intermetallic compounds, a total of 7 in the Au–Ce binary system and 9 in the Ce–Sn binary system, were primarily treated as line compounds. The Gibbs energy values per mole of formula unit Au_mCe_n and Ce_pSn_q were expressed as

$$G^{\mathrm{Au}_m\mathrm{Ce}_n} = mG_{Au}^{Fcc} + nG_{\mathrm{Ce}}^{Fcc} + \Delta G_f^{\mathrm{Au}_m\mathrm{Ce}_n} \tag{5}$$

and

$$G^{\mathrm{Ce}_p\mathrm{Sn}_q} = pG_{\mathrm{Ce}}^{Fcc} + qG_{\mathrm{Sn}}^{Bct} + \Delta G_f^{\mathrm{Ce}_p\mathrm{Sn}_q} \tag{6}$$

where $\Delta G_f^{\mathrm{Au}_m\mathrm{Ce}_n}$ and $\Delta G_f^{\mathrm{Ce}_p\mathrm{Sn}_q}$ are the Gibbs energy values of formation of per mole of formula unit of the Au_mCe_n and Ce_pSn_q, respectively. Due to lack of experimental information, it was assumed that the Neumann–K·opp rule applies to the heat capacity in both Au–Ce, and Ce–Sn systems, i.e., $\Delta C_p = 0$. Hence $\Delta G_f^{\mathrm{Au}_m\mathrm{Ce}_n}$ and $\Delta G_f^{\mathrm{Ce}_p\mathrm{Sn}_q}$ can be expressed as

$$\Delta G_f^{\mathrm{Au}_m\mathrm{Ce}_n} = A_1 + B_1T \tag{7}$$

and

$$\Delta G_f^{\mathrm{Ce}_p\mathrm{Sn}_q} = A_2 + B_2T \tag{8}$$

The constants A_i and B_i (i=1, 2) were assessed in the present work.

Out of the 16 binary compounds, 7 have been reported to exhibit significant ternary solubilities, and the other three to have a small ternary solubility (Table 4) on the isothermal section at 1023 K. Therefore, the sublattice model (Au, Sn)$_u$: (Ce)$_v$ was adopted here to describe the ternary solubility in Au_mCe_n and Ce_pSn_q phases. Again, the Neumann–Kopp rule is adopted due to the lack of experimental heat capacities for these phases. Their Gibbs energy of the compounds (Au, Sn)$_u$: (Ce)$_v$ can be expressed as follows:

$$\begin{aligned} G^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})_v} = {} & Y_{\mathrm{Au}}^I G_{\mathrm{Au:Ce}}^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})_v} + Y_{\mathrm{Sn}}^I G_{\mathrm{Sn:Ce}}^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})_v} \\ & + \frac{u}{u+v}RT(Y_{\mathrm{Au}}^I \mathrm{Ln}Y_{\mathrm{Au}}^I + Y_{\mathrm{Sn}}^I \mathrm{Ln}Y_{\mathrm{Sn}}^I) \\ & + Y_{\mathrm{Au}}^I Y_{\mathrm{Sn}}^I \Big(\sum_{j=0,1} {}^{(j)}L_{\mathrm{Au,Sn:Ce}}^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})^v}(Y_{\mathrm{Au}}-Y_{\mathrm{Sn}})^j\Big) \end{aligned} \tag{9}$$

here

$$G_{\mathrm{Sn:Ce}}^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})_v} = \frac{u}{u+v}{}^0G_{\mathrm{Sn}}^{Bct} + \frac{v}{u+v}{}^0G_{\mathrm{Ce}}^{Fcc} + E_1 + F_1T \tag{10}$$

$$G_{Au:Ce}^{(Au,Sn)_u:(Ce)_v} = \frac{u}{u+v}{}^0G_{Au}^{Fcc} + \frac{v}{u+v}{}^0G_{Ce}^{Fcc} + E_2 + F_2T \tag{11}$$

$$L_{\mathrm{Au,Sn:Ce}}^{(\mathrm{Au,Sn})_u:(\mathrm{Ce})_v} = E_3 + F_3T \tag{12}$$

where Y_{Au}^I, and Y_{Sn}^I are the site fractions of Au and Sn in the first sublattice of (Au, Sn)$_u$:(Ce)$_v$, and the parameters E_k and F_k (k=1, 2, 3) were optimized in this work.

3.2.3. Ternary compounds

In the ternary system, a total of 7 ternary compounds have been reported [18]. In this work they are described according to the crystal structure and homogeneity ranges reported in the literature. τ_1-$Ce_3Au_{13}Sn_4$ was modeled as a semi-stoichiometric phase ((Ce)$_{0.15}$:(Au, Sn)$_{0.85}$). τ_2-$CeAu_{2-x}Sn_x$, τ_3-$CeAu_2Sn$, τ_6-$CeAu_2Sn_2$, and τ_7-$Ce_2Au_3Sn_4$ were represented by three sublattice sites, i.e. Ce:$Au_{1.34}$:(Au,Sn)$_{0.66}$, $Ce_{0.25}$:$Au_{0.5}$:(Au,Sn)$_{0.25}$, $Ce_{0.2}$:(Au,Sn)$_{0.4}$:$Sn_{0.4}$, and $Ce_{0.222}$:$Au_{0.334}$:(Au,Sn)$_{0.444}$, respectively. The two remaining compounds, i.e. τ_4-CeAuSn and τ_5-Ce_5AuSn_3, were treated as stoichiometric phase $Au_xCe_ySn_z$. The Gibbs energy expressions of all these models were listed as follows.

The Gibbs energy expression for the semi-stoichiometric compound, τ_1, was written as

$$G^{Ce_{0.15}:(Au,Sn)_{0.85}} = Y^{II}_{Au}G^{\tau_1}_{Ce:Au} + Y^{II}_{Sn}G^{\tau_1}_{Ce:Sn} + 0.85RT(Y^{II}_{Au}\mathrm{Ln}Y^{II}_{Au} + Y^{II}_{Sn}\mathrm{Ln}Y^{II}_{Sn}) + Y^{II}_{Au}Y^{II}_{Sn}(\sum_{j=0,1..} {}^{j}L_{Ce:Au,Sn}(Y_{Au}-Y_{Sn})j) \quad (13)$$

where

$$G^{\tau_1}_{Ce:Au} = 0.15\,{}^0G^{Fcc}_{Ce} + 0.85\,{}^0G^{Fcc}_{Au} + H_1 + J_1T \quad (14)$$

$$G^{\tau_1}_{Ce:Sn} = 0.15\,{}^0G^{Fcc}_{Ce} + 0.85\,{}^0G^{Bct}_{Sn} + H_2 + J_2T \quad (15)$$

The $L_{Ce:Au,Sn}$ represented the interaction between Au and Sn in the second sublattice, expressed by Redlich–Kister polynomials:

$$L^{\tau_1}_{Ce:Au,Sn} = H_3 + J_3T \quad (16)$$

Gibbs energy expression for any of the stoichiometric compounds (i.e. τ_4 andτ_5) can be given as

$$G^{Au_xCe_ySn_z} = \frac{x}{x+y+z}{}^0G^{Fcc}_{Au} + \frac{y}{x+y+z}{}^0G^{Fcc}_{Ce} + \frac{z}{x+y+z}{}^0G^{Bct}_{Sn} + H_4 + J_4T \quad (17)$$

and for phases, τ_3 and τ_7 (Ce_x:Au_y:$(Au,Sn)_z$), it can be written in the following way:

$$G^{Ce_xAu_y(Au,Sn)_z} = Y^{III}_{Au}G_{Ce:Au:Au} + Y^{III}_{Sn}G_{Ce:Au:Sn} + \frac{z}{x+y+z}RT(Y^{III}_{Au}\mathrm{Ln}Y^{III}_{Au} + Y^{III}_{Sn}\mathrm{Ln}Y^{III}_{Sn}) + Y^{III}_{Au}Y^{III}_{Sn}(\sum_{j=0,1...} {}^0L_{Ce:Au:Au,Sn}(Y_{Au}-Y_{Sn})^j) \quad (18)$$

where

$$G_{Ce:Au:Au} = \frac{x}{x+y+z}{}^0G^{Fcc}_{Ce} + \frac{y+z}{x+y+z}{}^0G^{Fcc}_{Au} + H_5 + J_5T \quad (19)$$

$$G_{Ce:Au:Sn} = \frac{x}{x+y+z}{}^0G^{Fcc}_{Ce} + \frac{y}{x+y+z}{}^0G^{Fcc}_{Au} + \frac{z}{x+y+z}{}^0G^{Bct}_{Sn} + H_6 + J_6T \quad (20)$$

$$L_{Ce:Au:Au,Sn} = H_7 + J_7T \quad (21)$$

As for the τ_6 phase, its Gibbs energy expression is

$$G^{Ce_{0.2}:(Au,Sn)_{0.4}:Sn_{0.4}} = Y^{II}_{Au}G_{Ce:Au:Sn} + Y^{II}_{Sn}G_{Ce:Sn:Sn} + 0.4RT(Y^{II}_{Au}\mathrm{Ln}Y^{II}_{Au} + Y^{II}_{Sn}\mathrm{Ln}Y^{II}_{Sn}) + Y^{II}_{Au}Y^{II}_{Sn}(\sum_{j=0,1...} {}^jL_{Ce:Au,Sn::Sn}(Y_{Au}-Y_{Sn})^j) \quad (22)$$

where

$$G_{Ce:Au:Sn} = 0.2\,{}^0G^{Fcc}_{Ce} + 0.4\,{}^0G^{Fcc}_{Au} + 0.4\,{}^0G^{Bct}_{Sn} + H_8 + J_8T \quad (23)$$

$$G_{Ce:Sn:Sn} = 0.2\,{}^0G^{Fcc}_{Ce} + 0.8\,{}^0G^{Bct}_{Sn} + H_9 + J_9T \quad (24)$$

$$L_{Ce:Au,Sn::Sn} = H_{10} + J_{10}T \quad (25)$$

The parameters H_z and J_z (z =1, 2, … ,10) were optimized in this work.

4. Results and discussion

4.1. Ab initio calculations

The enthalpies of formation for a total of 17 compounds were calculated utilizing the *ab initio* approach. In other words, the enthalpies of formation of all the binary intermetallic compounds in Au–Ce and Ce–Sn systems (except for that of $Au_{51}Ce_{14}$), and 2 ternary compounds (τ_4 and τ_5) were calculated with this method. As shown in Table 4, these results are consistent with the data from the literature [17,19,34–37]. The deviation between the *ab initio* results and those reported in [17,19,34–37] is less than 13.5%. This type of deviation is acceptable when taking into consideration the following reasons: (a) the thermochemical data were obtained at different temperatures, i.e. 0 K for the *ab initio* approach, and 298 K/300 K for those reported in [17,19,34–37], and (b) there is always some degree of error in both of the experimentally determined and theoretically calculated values.

In the present work, the enthalpies of formation based on *ab initio* calculation were adopted as the initial thermodynamic parameters (e.g. values of A_i in Eqs. (7) and (8), as described in Section 3.2) for the compounds in question. Figs. 1 and 2 illustrate all the thermodynamic values including the data from the previous reports [17,19,34–37], and the one obtained in this work (i.e. both the *ab initio* calculated values and the thermodynamically assessed data).

4.2. Thermodynamic assessment

Thermodynamic parameters were optimized in this work by using the PARROT module in the Thermo-Calc software (TCCS) as shown in Table 8. The optimization was carried out by using all the experimental and theoretical data available for all the three binary

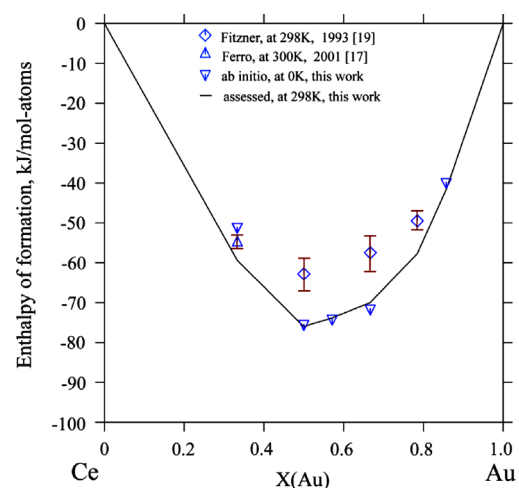

Fig. 1. Enthalpies of formation of compounds in Au–Ce binary system, combining with data from former investigations. Reference states: Au (fcc), and Ce (fcc).

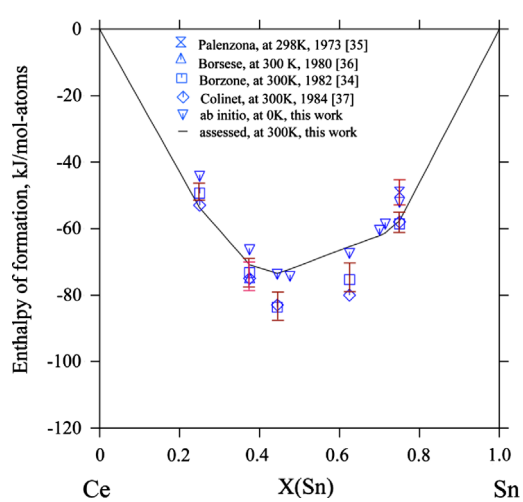

Fig. 2. Enthalpies of formation of compounds in Ce–Sn binary system, combining with data from former investigations. Reference states: Ce (fcc) and Sn (bct).

Table 8
Thermodynamic parameters in Au–Ce–Sn ternary system, used in this work.

Phase	Thermodynamic parameters
Liquid model: (Au, Ce, Sn)	${}^{0}G^{Liq}_{Au,Sn} = -48822.94 + 19.6767T$
	${}^{1}G^{Liq}_{Au,Sn} = -17990.36 + 0.7892T$
	${}^{2}G^{Liq}_{Au,Sn} = -5222.53$
	${}^{0}G^{Liq}_{Au,Ce} = -25446.238 + 6.002T$
	${}^{1}G^{Liq}_{Au,Ce}1 = -85046.070$
	${}^{2}G^{Liq}_{Au,Ce} = +7400$
	${}^{0}G^{Liq}_{Ce,Sn} = -209129.135 + 20T$
	${}^{1}G^{Liq}_{Ce,Sn} = +27189.0474$
	${}^{2}G^{Liq}_{Ce,Sn} = +10000$
Bcc model: (Au,Ce,Sn)	${}^{0}G^{Bcc}_{Ce,Sn} = -124000-20.0580T$
Fcc model: (Au,Ce,Sn)	${}^{0}G^{Fcc}_{Au,Sn} = -28802.84-5.5753T$
	${}^{1}G^{Fcc}_{Au,Sn} = +8815.07-9.7873T$
Hcp model: (Au,Ce,Sn)	${}^{0}G^{Hcp}_{Au,Sn} = -13163.983-4.4859T$
	${}^{0}G^{Hcp}_{Au,Sn} = -18954.810-23.7973T$
$AuCe_2$ model: $(Au)_{0.3333}$:$(Ce)_{0.6667}$	$G^{AuCe_2}_{Au:Ce} = -59392.7618 + 7.3464T + 0.3333GHSERAU + 0.6667GHSERCE$
βAuCe model: $(Au,Sn)_{0.5}$:$(Ce)_{0.5}$	$G^{\beta AuCe}_{Au:Ce} = -75701.3423 + 0.8103T + 0.5GHSERAU + 0.5GHSERCE$
	$G^{\beta AuCe}_{Sn:Ce} = +133560-1149.5T + 141.3834T\mathrm{Ln}(T) + 0.5GHSERSN + 0.5GHSERCE$
	$L^{\beta AuCe}_{Au,Sn:Ce} = +111540-1054.3T + 130.6928T\mathrm{Ln}(T)$
αAuCe model: $(Au)_{0.5}$:$(Ce)_{0.5}$	$G^{\alpha AuCe}_{Au:Ce} = -76017.6953 + 1.2303T + 0.5GHSERAU + 0.5GHSERCE$
Au_4Ce_3 model: $(Au)_{0.5714}$:$(Ce)_{0.4286}$	$G^{Au_4Ce_3}_{Au:Ce} = -73820 + 1.145T + 0.5714GHSERAU + 0.4286GHSERCE$
Au_2Ce model: $(Au,Sn)_{0.6667}$: $(Ce)_{0.3333}$	$G^{Au_2Ce}_{Au:Ce} = -70000 + 0.28T + 0.6667GHSERAU + 0.3333GHSERCE$
	$G^{Au_2Ce}_{Sn:Ce} = -24686.679-410.7384T + 58.1188T\mathrm{Ln}(T) + 0.6667GHSERSN + 0.3333GHSERCE$
	${}^{0}L^{Au_2Ce}_{Au,Sn:Ce} = -38862.116-548.619T + 74.2041T\mathrm{Ln}(T)$
$Au_{51}Ce_{14}$ model: $(Au, Sn)_{0.7846}$:$(Ce)_{0.2154}$	$G^{Au_{51}Ce_{14}}_{Au:Ce} = -57709.608 + 0.6668T + 0.7846GHSERAU + 0.2154GHSERCE$
	$G^{Au_{51}Ce_{14}}_{Sn:Ce} = -35503.995 + 10.7278T + 0.7846GHSERSN + 0.2154GHSERCE$
	${}^{0}L^{Au_{51}Ce_{14}}_{Au,Sn:Ce} = -66675.7984 + 7.3566T$
Au_6Ce model: $(Au)_{0.8571}$:$(Ce)_{0.1429}$	$G^{Au_6Ce}_{Au:Ce} = -41593.6344 + 2.6582T + 0.8571GHSERAU + 0.1429GHSERCE$
$Au_{10}Sn$_Betal model: (Au,Sn)	${}^{0}G^{Au_{10}Sn_Betal}_{Au} = +125 + 0.79T + GHSERAU$
	${}^{0}G^{Au_{10}Sn_Betal}_{Sn} = +3804-3.46T + GHSERSN$
	${}^{0}L^{Au_{10}Sn_Betal}_{Au,Sn} = -13546.6569-4.7107T$
	${}^{0}L^{Au_{10}Sn_Betal}_{Au,Sn} = -14193.13-22.3922T$
Au_5Sn model: $(Au)_{0.84}$: $(Sn)_{0.16}$	$G^{Au_5Sn}_{Au:Sn} = -4036.7822-3.3646T + 0.84GHSERAU + 0.16GHSERSN$
$AuSn_2$ model: $(Au)_{0.3333}$:$(Sn)_{0.6667}$	$G^{AuSn_2}_{Au:Sn} = -14296.5524-5.4538T + 0.3333GHSERAU + 0.6667GHSERSN$

Table 8 (*continued*)

Phase	Thermodynamic parameters
$AuSn_4$_Etal model: $(Au)_{0.2}$:$(Sn)_{0.8}$	$G^{AuSn_4_Etal}_{Au:Sn} = -7790.257 + 0.8355T + 0.2GHSERAU + 0.8GHSERSN$
AuSn_Delta model: $(Sn)_1$:$(Sn,Va)_1$: $(Au)_1$	$G^{AuSn_Delta}_{Sn:Va:Au} = -31490 + 4.3439T + GHSERAU + GHSERSN$
	$G^{AuSn_Delta}_{Sn:Sn:Au} = -1000 + 0.5T + GHSERAU + 2GHSERSN$
	${}^{0}L^{AuSn_Delta}_{Sn:Sn,Va:Sn} = -20000 + T$
Ce_3Sn model: $(Ce)_{0.75}$: $(Sn)_{0.25}$	$G^{Ce_3Sn}_{Ce:Sn} = -53717.004 + 8.7753T + 0.25GHSERSN + 0.75GHSERCE$
Ce_5Sn_3 model: $(Ce)_{0.625}$:(Au, $Sn)_{0.375}$	$G^{Ce_5Sn_3}_{Ce:Sn} = -70360.3499 + 6.4692T + 0.375GHSERSN + 0.625GHSERCE$
	$G^{Ce_5Sn_3}_{Ce:Au} = -28100 + 0.375GHSERAU + 0.625GHSERCE$
	${}^{0}L^{Ce_5Sn_3}_{Ce:Au,Sn} = -28000$
Ce_5Sn_4 model: $(Ce)_{0.5556}$:(Au, $Sn)_{0.4444}$	$G^{Ce_5Sn_4}_{Ce:Sn} = -73463.2507 + 5.7042T + 0.4444GHSERSN + 0.5556GHSERCE$
	$G^{Ce_5Sn_4}_{Ce:Au} = -31824.9668 + 1.0201T + 0.4444GHSERAU + 0.5556GHSERCE$
	${}^{0}L^{Ce_5Sn_4}_{Ce:Au,Sn} = -64696.3068 + 13.1841T$
$Ce_{11}Sn_{10}$ model: $(Ce)_{0.5238}$:$(Sn)_{0.4762}$	$G^{Ce_{11}Sn_{10}}_{Ce:Sn} = -72205.5443 + 4.9042T + 0.4762GHSERSN + 0.5238GHSERCE$
Ce_3Sn_5 model: $(Ce)_{0.375}$:(Au, $Sn)_{0.625}$	$G^{Ce_3Sn_5}_{Ce:Sn} = -65379.7835 + 2.9979T + 0.625GHSERSN + 0.375GHSERCE$
	$G^{Ce_3Sn_5}_{Ce:Au} = -63679.515 + 9.7461T + 0.625GHSERAU + 0.375GHSERCE$
	${}^{0}L^{Ce_3Sn_5}_{Ce:Au,Sn} = -86374.872 + 42.2042T$
Ce_3Sn_7 model: $(Ce)_{0.3}$: $(Sn)_{0.7}$	$G^{Ce_3Sn_7}_{Ce:Sn} = -62186.033 + 3.8709T + 0.7GHSERSN + 0.3GHSERCE$
Ce_2Sn_5 model: $(Ce)_{0.2857}$:$(Sn)_{0.7143}$	$G^{Ce_2Sn_5}_{Ce:Sn} = -61297.117 + 3.8567T + 0.7143GHSERSN + 0.2857GHSERCE$
$CeSn_3$ model: $(Ce)_{0.25}$: $(Au,Sn)_{0.75}$	$G^{CeSn_3}_{Ce:Sn} = -57620.424 + 3.1225T + 0.75GHSERSN + 0.25GHSERCE$
	$G^{CeSn_3}_{Ce:Au} = -54283.419 + 20.8409T + 0.75GHSERAU + 0.25GHSERCE$
	${}^{0}L^{CeSn_3}_{Ce:Sn} = --87410.714 + 27.3810T$
τ_1 model: $(Ce)_{0.15}$:(Au, $Sn)_{0.85}$	$G^{\tau_1}_{Ce:Au} = -40800 + 2.01T + 0.85GHSERAU + 0.15GHSERCE$
	$G^{\tau_1}_{Ce:Sn} = -13455 + 2.06T + 0.85GHSERSN + 0.15GHSERCE$
	${}^{0}L^{\tau_1}_{Ce:Au,Sn} = -85000 + 3.06T$
τ_2 model: (Ce):$(Au)_{1.34}$: $(Au,Sn)_{0.66}$	$G^{\tau_2}_{Ce:Au:Sn} = -260000 + 10.032T + 1.34GHSERAU + GHSERCE + 0.66GHSERSN$
	$G^{\tau_2}_{Ce:Au:Au} = -197900 + 2GHSERAU + GHSERCE$
	${}^{0}L^{\tau_2}_{Ce:Au:(Au,Sn)} = -16000$

Table 8 (*continued*)

Phase	Thermodynamic parameters
τ_3 model: $(Ce)_{0.25}$: $(Au)_{0.5}$:$(Au,Sn)_{0.25}$	$G^{\tau_3}_{Ce:Au:Sn} = -78400 + 3.08T + 0.5GHSERAU + 0.25GHSERCE + 0.25GHSERSN$
	$G^{\tau_3}_{Ce:Au:Au} = -29000 + 2.02T + 0.75GHSERAU + 0.25GHSERCE$
	${}^0L^{\tau_3}_{Ce:Au:Au,Sn} = --36900 + 2.02T$
τ_4 model: (Ce):(Au): (Sn)	$G^{\tau_4}_{Ce:Au:Sn} = -259400 + 8.032T + GHSERAU + GHSERCE + GHSERSN$
τ_5 model: $(Ce)_5$:(Au): $(Sn)_3$	$G^{\tau_5}_{Ce:Au:Sn} = -673000 + 24.9196T + GHSERAU + 5GHSERCE + 3GHSERSN$
τ_6 model: $(Ce)_{0.2}$: $(Au,Sn)_{0.4}$:$(Sn)_{0.4}$	$G^{\tau_6}_{Ce:Au:Sn} = -71000 + 4.05T + 0.4GHSERAU + 0.2GHSERCE + 0.4GHSERSN$
	$G^{\tau_6}_{Ce:Sn:Sn} = -12000 + 2T + +0.2GHSERCE + 0.8GHSERSN$
	${}^0L^{\tau_6}_{Ce:Au,Sn:Sn} = -24000$
τ_7 model: $(Ce)_{0.222}$: $(Au)_{0.334}$: $(Au,Sn)_{0.444}$	$G^{\tau_7}_{Ce:Au:Sn} = -71900 + 3.05T + 0.334GHSERAU + 0.222GHSERCE + 0.444GHSERSN$
	$G^{\tau_7}_{Ce:Au:Au} = -19000 + 0.5T + 0.222GHSERCE + 0.778GHSERAU$
	${}^0L^{\tau_7}_{Ce:Au:Au,Sn} = -48000$

systems, as well as those for the Au–Ce–Sn ternary system. As discussed in the following sections, the present calculations can reproduce most of the experimentally determined and theoretically calculated data of all the constituent systems.

4.2.1. Au–Ce binary system

The thermodynamic assessment of the Au–Ce binary system was carried out in the present work, complemented with the *ab initio* derived enthalpies of formation of the relevant IMCs, as well as the thermochemical data available in the literature [17,19]. The compositions and temperatures of the invariant reactions in the binary system are listed in Table 2, including both the experimental data [9,11,17,18] and the assessed values [13,14,16]. A comparison was made, namely, between the phase diagram assessed in this work and the data from previous works [9,11,13,14,16–18]. As shown in Fig. 3, the present calculations fit well with those experimentally determined points [9,11] with only exceptions: (a) the temperature of eutectic reaction involving βAuCe and Au_2Ce phases being 1337 K, instead of 1268 K from [9,13,18], or 1353 K from [14]; (b) the eutectic point being closer to the Au-rich side here than in [9,13,18]; and (c) the slight deviation of the phase boundary (i.e. involving βAuCe and $AuCe_2$ at Ce-rich side) between the work of [9] and the present calculated results. The present thermodynamic description can be considered acceptable due to the following reasons:

(i) At a certain temperature, comparing the width of the two-phase fields on each side of a given compound, judgments about the asymmetry of the phase can be derived [29,30]. If the width of the two-phase field on one side of the compound is either one-half or double that on the other side, the asymmetry may be considered to be unusual. Regarding Au_2Ce phase, the ratio of these widths is definitely more than two times in the phase diagram reported previously [9,13,18]. Thus, the description of that part of the diagram must be improved. In order to fulfill both the liquidus symmetry and the enthalpy of formation of the Au_2Ce, the eutectic temperature (1268 K) had to be sacrificed, similarly to that done in [14]. However, it is to be noted that the present eutectic temperature is much closer to the experimental result than the one assessed in [14].

(ii) On the other hand, if the liquidus asymmetry of the Au_2Ce in Ref. [13] can be considered to be reasonable, it implies that this phase is unstable at low temperature (probably as high as 1273 K [38]), or atomic clustering should occur in the liquid state. However, no experimental information indicates the decomposition of this phase at low temperature. Consequently, the liquid should then be treated as an associate-solution rather than the regular-solution. By utilizing the associate model, the liquidus can in fact be fitted better with the experimental data than by utilizing the regular solution model. However, according to the criterion of the "associates" the enthalpy of mixing curve of the system should have a pronounced "V"-shape [39]. According to the mixing enthalpies reported in [19], there is no such shape. Therefore, it can be concluded that the associate-solution is perhaps not a suitable model for the present occasion.

(iii) Just as discussed in Section 2.1, the deviation of liquidus in Ce-rich side is acceptable due to the accuracy issues. Besides, an asymmetry liquidus of βAuCe was proposed in [9], i.e. ratio of widths: 6.5/3.

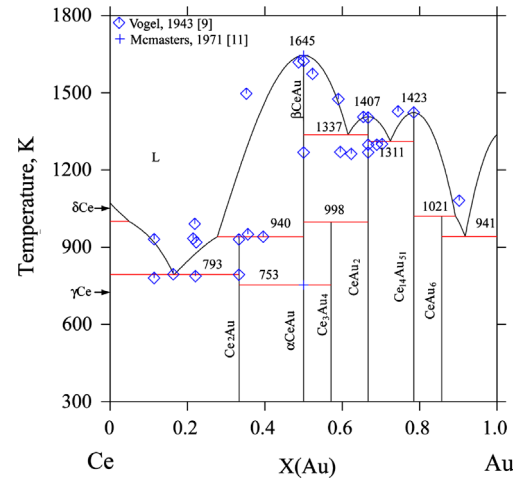

Fig. 3. Phase diagram of Au–Ce binary system, comparing with experimentally determined values [9,11].

The issues related to the uncertain reaction temperature involving the Au_4Ce_3 and Au_6Ce phases were already discussed in Section 2.1. Based on the ternary information, a slightly lower temperature than 1023 K was adopted for the formation of both Au_4Ce_3 and Au_6Ce in this assessment, as shown in Table 2.

When the thermochemical information of Au–Ce system is considered, the present thermodynamic parameters are consistent with the experimentally determined mixing enthalpies at 1473 K [19], as well as with the experimentally measured enthalpies of formation of IMCs at around 298 K. As shown in Fig. 1, the assessed standard enthalpies of formation agreed with both the *ab initio* approach calculated values and the experimentally measured data [17,19]. As shown in Fig. 4, the assessed enthalpies of mixing match well with the experimental data given by the high-temperature calorimetry at 1473 ± 2 K [19].

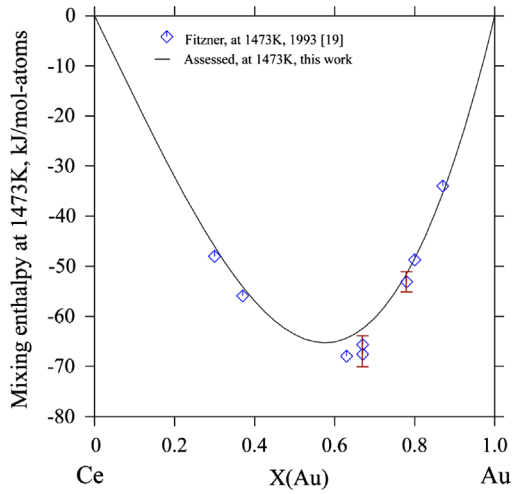

Fig. 4. Mixing enthalpies of Au–Ce binary system at 1473 K, combining with data from experimental investigations. Reference states: Au (liquid) and Ce (liquid).

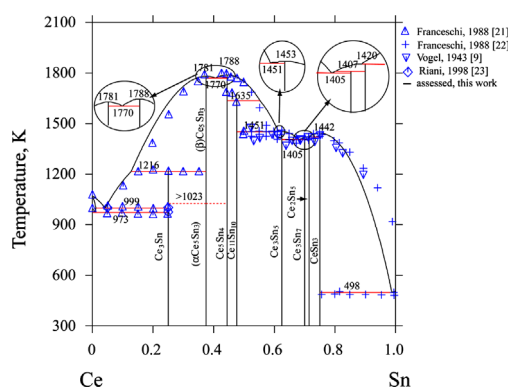

Fig. 5. Phase diagram of Ce–Sn binary system, comparing with experimentally determined values.

4.2.2. Ce–Sn binary system

This binary Ce–Sn system is more complex than the above described Au–Ce system. The Ce–Sn system involves 8 compounds and 13 invariant reactions. As shown in Fig. 5 and Table 3, the calculated results using the current thermodynamic description match well with the experimental information, except for three issues. (i) First exception is the deviation of liquidus boundary of L +Ce_3Sn at the Sn-rich side between the experimental information and the calculated result. (ii) Formation mechanism of the Ce_5Sn_3 phase is different. Specifically, in this work, the Ce_5Sn_3 phase was treated to be the product from a congruent reaction (L→Ce_5Sn_3) rather than from the peritectic reaction (L+Ce_5Sn_4→Ce_5Sn_3) [21]. In addition an extra eutectic reaction L→Ce_5Sn_3+Ce_5Sn_4 was introduced. (iii) The second exception is about the eutectic temperature of the reaction L→$Ce_{11}Sn_{10}$+Ce_3Sn_5. In this work, 1451 K was employed instead of the experimentally determined 1428 ± 10 K [22]. Again these deviations are acceptable when taking the following reasons into account.

For the liquidus boundary, an excessive slope change was proposed around 920 K in the experimental work, as shown in Fig. 5. Generally, such change will be referred to as a polymorphic transformation at that temperature. However, this type of reaction was not discovered. Therefore, a flatter liquidus boundary should be considered, just as that has been done in the present calculation.

With regard to the reaction type, an unusually asymmetric liquidus (ratio of widths: 1/0.2) for the Ce_5Sn_4 phase was proposed in the experimentally determined phase diagram [22]. However, no evidence for the decomposition of Ce_5Sn_4 at low temperature was shown. In addition, the ternary thermodynamic description is simplified by excluding the associated model for the liquid phase. Therefore, the required increase in the symmetry of the liquidus was to be achieved by other means. In order to obtain the symmetric liquidus, two methods can be attempted, by either decreasing the two phase range (L+Ce_5Sn_4) at the Ce-rich side, or by increasing the two phase range at the Sn-rich side. The former solution would lead to the shift in the composition of liquid of the proposed peritectic reaction towards the Sn-rich side. Hereby, this method would introduce a new problem whether it should

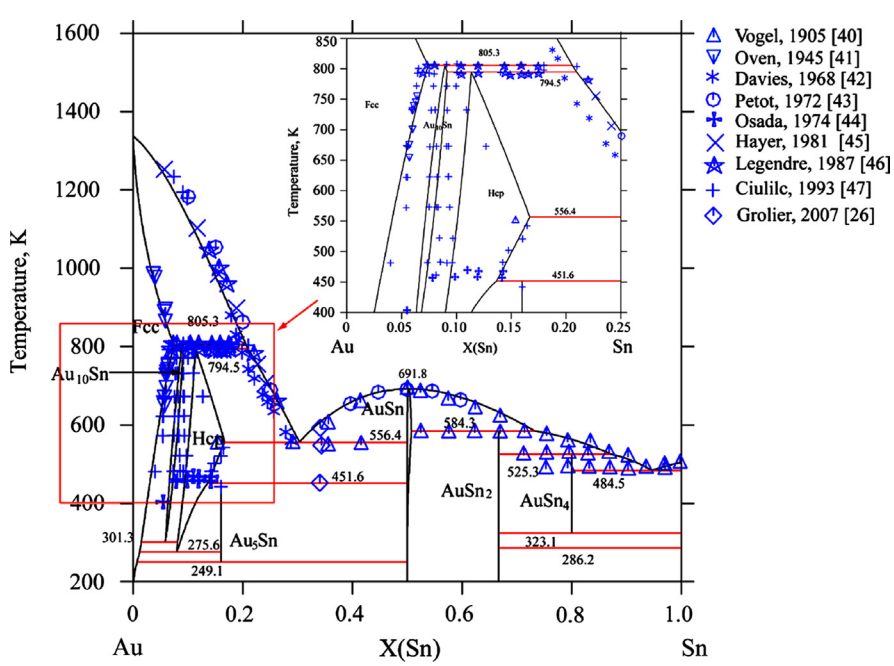

Fig. 6. Phase diagram of Au–Sn binary system as calculated with the present thermodynamic description compared with experimental data [26,40–47].

maintain the peritectic reaction type or the reaction temperature. If the peritectic reaction type was to be maintained, the reaction temperature would have to be significantly sacrificed, and vice versa. The second solution, increasing the width of L+Ce_5Sn_4 at Sn-rich side, will suppress the eutectic reaction (L→$Ce_{11}Sn_{10}$+Ce_3Sn_5), and will change the formation mechanism of Ce_3Sn_5. Thus, if this method was used, it would lead to a larger deviation from the determined phase diagram [22]. Therefore, the best solution here was to decrease the L+Ce_5Sn_4 range at the Ce-rich side by keeping the reported reaction temperature, in order to reproduce more experimental values.

When considering the discrepancy of the calculated eutectic temperature from that of the experimentally determined one [22], it is as a result of the different steepness of the liquidus boundaries relating the $Ce_{11}Sn_{10}$ and Ce_3Sn_5 phases in the proposed phase diagram [22]. As stated by Okamoto [38], *"if two line compounds in a binary system are stable to low temperatures, the steepness of their liquidus boundaries must be comparably similar."* Hereby, the liquidus boundary of $Ce_{11}Sn_{10}$ had to be modified to be flatter. Thus the temperature of the involved eutectic reaction became higher than the reported one [18].

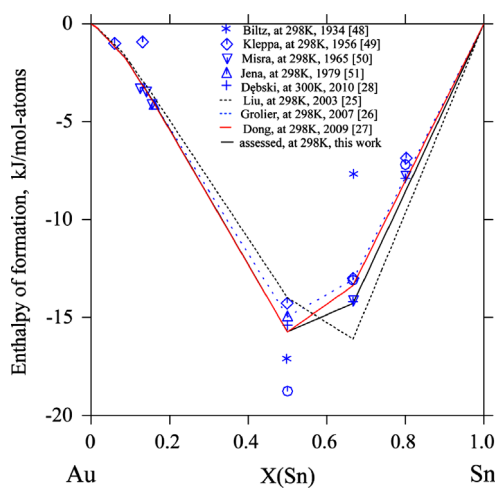

Fig. 7. Standard enthalpy of formation of Au–Sn system; comparison between the experimentally determined data [48–51], the calculated results using parameters from [25–27] and with the present thermodynamic description. Reference states: Au(fcc) and Sn(bct).

The thermodynamic properties of the Ce–Sn system are shown in Table 4, and Fig. 2. The comparison was carried out between the present work (i.e. the assessed values, and the *ab initio* calculated data) and the results reported in [34–37]. As can be seen, the presented calculated results are in coincidence with those former reported information.

4.2.3. Au–Sn binary system

The thermodynamic description of the Au–Sn binary system was modified in the present work by using the recently reported new lattice stability of Sn(hcp) by SGTE on the basis of the experimentally determined information [26,40–49]. As shown in Fig. 6 and Table 5, the experimental determined phase diagram can be reproduced by using the present parameters. Besides, the standard enthalpies of formation of this system are illustrated in Fig. 7. As shown in Fig. 7, it is a large deviation between the abnormal asymmetric curve calculated from the parameters of Liu et al. [25] and these experimentally measured data. This deviation is one of the main reasons prompting the reassessment work of [26] or the modification shown in [27]. The currently assessed standard enthalpies of formation fit with the experimentally determined values [48–51] and the work of [26], except for the large deviation from the values determined by Biltz et al. [48], the *ab initio* approach calculated enthalpy of formation of AuSn from [27], and one point from the work of [49]. Nevertheless, in the present work, these values were given a smaller weight to the values in [48] than the others since the approach employed by Biltz et al. [48], i.e. aqueous solution calorimetry, was not regarded as a "high-accuracy" method. In [27], the authors have already analyzed the accuracy of the *ab initio* approach calculated enthalpy of formation of AuSn. Thus these data are also given a smaller weight than the remaining data. Besides, just as analyzed in [26], the deviation between the present calculated result and one of the experimental point of Kleppa et al. [49] can be accepted. Hence, the parameters obtained in this work are acceptable, and can be used to extrapolate multi-component system.

4.2.4. Au–Ce–Sn ternary system

This ternary system was thermodynamically assessed based upon the reported isothermal section at 1023 K [18] and the data concerning the three binary systems. As shown in Fig. 8b, the present thermodynamic description can reproduce the experimentally determined phase relations, only with exceptions of some tie lines of βAuCe and τ_1 phases. Comparing Fig. 8a and b, the proposed

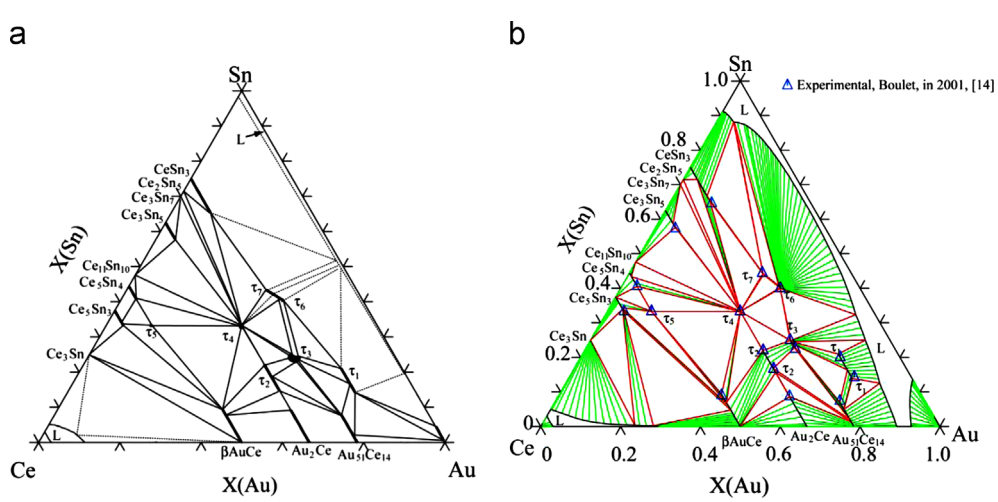

Fig. 8. Isothermal section of Au–Ce–Sn ternary system at 1023 K: (a) a schematic diagram of the proposed isothermal [12]; and (b) the calculated isothermal section in this work, with experimental information superimposed.

tie line of Ce_3Sn+βAuCe was replaced by that of Ce_5Sn_3+liquid in this work, and the proposed tie lines of τ_6+τ_1, and τ_1+Au were suppressed by those of τ_3+liquid, and τ_1+liquid, respectively. These conflictions are acceptable, since the liquid phase should have a larger range than that proposed in isothermal section according to the phase diagram of Au–Ce and Ce–Sn systems. And with the introduction of such a large range of liquid phase, the nearby phase equilibria would be affected. Furthermore, the calculated ternary solubilities of binary compounds as well as the homogeneity ranges of ternary compounds are identical with the experimental information (Table 7), except for the range of τ_3. In this work, τ_3 was treated as a compound with the composition exchanging between Au and Sn rather than all the three elements illustrated in the proposed isothermal section [18]. The reason is that, in addition to homogeneity domain in the isothermal section, a formula of τ_3 (i.e. $CeAu_{2+x}Sn_{1-x}$ with $0<x<0.1$) was also suggested in [18]; thus a sample model was chosen in this work. Moreover, the liquidus projection was extrapolated by using the present thermodynamic parameters, as shown in Fig. 9. In addition, the ternary invariant reactions are listed in Table 9.

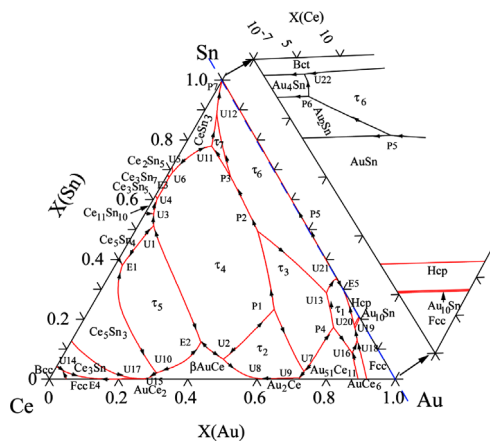

Fig. 9. Extrapolated liquidus projection of Au–Ce–Sn ternary system.

Table 9
Calculated invariant reactions in Au–Ce–Sn system.

Type	Reaction	Temperature, K
E1	$L\rightarrow\tau_5+Ce_5Sn_4+Ce_5Sn_3$	1765.72
E2	$L\rightarrow\tau_5+\beta AuCe+\tau_4$	1621.4
E3	$L\rightarrow\tau_4+Ce_3Sn_5+Ce_3Sn_7$	1400.9
E4	$L\rightarrow Fcc+Ce_3Sn+AuCe_2$	792.9
E5	$L\rightarrow\tau_1+AuSn+Hcp$	556.1
U1	$L+\tau_5\rightarrow\tau_4+Ce_5Sn_4$	1627.9
U2	$L+\tau_4\rightarrow\tau_2+\beta AuCe$	1623.8
U3	$L+Ce_5Sn_4\rightarrow\tau_4+Ce_{11}Sn_{10}$	1563.0
U4	$L+\tau_4\rightarrow Ce_{11}Sn_{10}+Ce_3Sn_5$	1452.4
U5	$L+CeSn_3\rightarrow\tau_4+Ce_2Sn_5$	1410.0
U6	$L+Ce_2Sn_5\rightarrow\tau_4+Ce_3Sn_7$	1401.0
U7	$L+\tau_3\rightarrow\tau_2+Au_{51}Ce_{14}$	1387.5
U8	$L+\tau_2\rightarrow Au_2Ce+\beta AuCe$	1338.5
U9	$L+\tau_2\rightarrow Au_{51}Ce_{14}+Au_2Ce$	1315.4
U10	$L+\tau_5\rightarrow Ce_5Sn_3+\beta AuCe$	1310.9
U11	$L+\tau_4\rightarrow\tau_7+CeSn_3$	1277.4
U12	$L+\tau_7\rightarrow CeSn_3+\tau_6$	1005.2
U13	$L+\tau_3\rightarrow\tau_6+\tau_1$	998.8
U14	$L+Bcc\rightarrow Fcc+Ce_3Sn$	972.6
U15	$L+\beta AuCe\rightarrow AuCe_2+Ce_5Sn_3$	935.8
U16	$L+Au_{51}Ce_{14}\rightarrow\tau_1+Au_6Ce$	934.2
U17	$L+Ce_5Sn_3\rightarrow Ce_3Sn+AuCe_2$	929.7
U18	$L+Au_6Ce\rightarrow\tau_1+Fcc$	793.6
U19	$L+Fcc\rightarrow\tau_1+Au_{10}Sn$	750.9
U20	$L+Au_{10}Sn\rightarrow\tau_1+Hcp$	740.1
U21	$L+\tau_6\rightarrow\tau_1+AuSn$	598.8
U22	$L+\tau_6\rightarrow Bct+AuSn_4$	484.4
P1	$L+\tau_2+\tau_4\rightarrow\tau_3$	1712.4
P2	$L+\tau_3+\tau_4\rightarrow\tau_6$	1517.7
P3	$L+\tau_4+\tau_6\rightarrow\tau_7$	1400.2
P4	$L+\tau_3+Au_{51}Ce_{14}\rightarrow\tau_1$	1075.8
P5	$L+\tau_6+AuSn\rightarrow AuSn_2$	584.3
P6	$L+\tau_6+AuSn_2\rightarrow AuSn_4$	525.6
P7	$L+\tau_6+CeSn_3\rightarrow Bct$	503.6

5. Conclusions

Enthalpies of formation of nearly all IMCs in Au–Ce and Ce–Sn systems at 0 K as well as for the two ternary compounds (i.e. τ_4 and τ_5) have been calculated by *ab initio* approach in this paper. The Au–Ce and Ce–Sn binary systems were optimized by utilizing the CALPHAD approach, and the previous description of the Au–Sn system was improved. Consequently, the phase relations in Au–Ce–Sn ternary system were thermodynamically assessed on the basis of the experimental data as well as the new descriptions of the three binary systems. Finally, a self-consistent set of thermodynamic parameters has been obtained. It can reliably reproduce most of the experimental information for the Au–Ce–Sn system.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.07.013.

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